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SPACE SYSTEMS COMMAND

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Safety

EXPLOSIVE YIELD DETERMINATION PROCESS FOR SPACE LAUNCH VEHICLES

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This handbook implements Department of the Air Force Instruction (DAFI) 91-202, *The Department of the Air Force (DAF) Mishap Prevention Program*. This handbook serves as a reference for implementing a consistent and data-driven approach to explosive yield determination for space launch vehicles. This handbook is applicable to HQ Space Systems Command (SSC), subordinate units and members, the Air National Guard, the Air Force Reserve Command, and customers at SSC spaceports. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with (IAW) the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the DAF Form 847, R*ecommendation for Change of Product*; route DAF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication for coordination prior to certification and approval.

Chapter 1—INTRODUCTION **3**

1.1. Purpose. 3

1.2. Applicability. 3

1.3. Background. 4

1.4. The Challenge of Novel Propellant Combinations. 4

1.5. The Importance of Managing Risk. 4

1.6. Relationship to Other Regulations and Directives. 4

Chapter 2—KEY CONCEPTS AND CONSIDERATIONS **5**

2.1. Defining Explosive Yield. 5

2.2. Traditional Propellants vs. Novel Propellants. 5

2.3. Factors Influencing Explosive Yield, Bounding Cases and Key Design Attributes. 5

2.4. Main Areas of Study. 5

2.5. The Role of Testing and Modeling. 6

2.6. Historical Context: Project PYRO 6

2.7. Blast Damage Assessment. 6

2.8. Other Key Considerations. 6

Chapter 3—INITIATING A PROPELLANT ASSESSMENT **7**

3.1. Initiation of Propellant Assessment. 7

3.2. Assessment Activities. 7

Chapter 4—THE THREE-STEP PROCESS: A DETAILED GUIDE **8**

4.1. Process. 8

4.2. Step 1: Theoretical-Analytic Review and Analysis. 8

4.3. Step 2: Characterization and Bounding Tests. 10

4.4. Step 3: Refinement Tests. 15

Chapter 5—CONCLUSION **17**

5.1. Key Takeaways. 17

5.2. Process Summary. 17

Chapter 6—ADDITIONAL TECHNICAL RESOURCES **18**

6.1. Recommendations for Further Reading. 18

6.2. Disclaimer. 18

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION **19**

Chapter 1

INTRODUCTION

1.1. Purpose. This handbook provides a blast damage assessment guide for space launch vehicles at Department of the Air Force launch complexes and a comprehensive guide to the scientifically-valid explosive yield determination process for space launch vehicles (LV), as outlined in the National Defense Authorization Act (NDAA) for Fiscal Year (FY) 2026, Section 1610 and FY24, Section 1601, paragraph b, respectively. This guide is intended to facilitate a standardized approach to assessing the risks associated with both traditional and novel propellant combinations. It is guidance for the user, which can be modified to fit the circumstances without a waiver. By following the methodology outlined in this handbook, users ensure that the resulting explosive yield determinations are defensible, reproducible, and objective, thereby meeting the mandated standards for a scientifically valid process. A range user may propose an alternative methodology and process for consideration and approval by SSC/SE.

1.2. Applicability.

1.2.1. This handbook is intended for individuals and organizations involved in space launch activities, including but not limited to: launch service providers, safety engineers, range safety personnel, government agencies (e.g., the Federal Aviation Administration [FAA], the National Aeronautics and Space Administration [NASA], the Department of the Air Force [DAF]), university researchers, payload providers, space flight participants, and international partners involved in collaborative launch efforts. The processes described in this handbook are applicable for new/novel propellants and existing propellants used in new ways.

1.2.2. The successful execution of the processes described herein relies on the coordination of three primary functional roles:

1.2.2.1. Subject Matter Experts (SMEs). Technical authorities specializing in, for example, propulsion chemistry, thermodynamics, fluid dynamics, and blast effects analysis. SMEs are critical for the theoretical-analytic reviews and ensuring the scientific integrity of the predictive models.

1.2.2.2. Test Personnel. The test teams responsible for the physical execution of lab-scale, small-scale, and large-scale tests. This includes, for example, range operators, instrumentation technicians, and data collection specialists who ensure all measurements are traceable and calibrated.

1.2.2.3. Safety Approval Authorities. The designated DAF officials (e.g., Chief of Safety, Range Safety Commander, SSC Director of Safety, SSC Commander) with the delegated authority to review the assessment findings, validate the hazard models, and formally accept the safety risks for launch operations.

1.2.3. Interagency Collaboration. The processes and methodologies detailed in this handbook are the product of deliberate collaboration between the DAF, FAA, and NASA. This interagency approach ensures the harmonization of safety standards across federal launch ranges, reduces redundant testing burdens for commercial providers, and facilitates the continuous exchange of data to refine explosive yield models.

1.2.4. This handbook applies exclusively to liquid propellants. Solid propellants are governed by existing Department of Transportation (DOT) hazard classification processes, which are leveraged to determine storage, handling, and explosive risks. Because solid propellants permit highly reliable hazard extrapolation from historically characterized formulations, additional testing is rarely required; however, should a novel solid formulation require unique hazard verification, the testing burden and cost shall be levied upon the manufacturer.

1.3. Background. The NDAA for FY26 Section 1610, mandated that the Secretary of the Air Force develop and publish a blast damage assessment guide for space launch vehicles at Air Force launch complexes. Additionally, legislative background explained how this guide should be developed consistent with NDAA FY24, Section 1601, paragraph b, which directed the establishment of a process through which scientifically-valid yield determinations can be assessed for space launch vehicles while in flight. This directive recognizes the evolving landscape of space launch activities, characterized by the ingress of commercial launch service providers and a corresponding increase in new launch vehicle designs that call for novel propellant use.

1.4. The Challenge of Novel Propellant Combinations. Public law, federal and internal directives require government entities to minimize the risk to the public, involved personnel, and critical assets. However, the space launch community cannot simply extrapolate explosive yield predictions for novel propellant combinations from experience with other propellants that are not chemically and thermodynamically similar. Not only is accounting for these differences analytically not viable, but the relatively small set of existing test data for other propellant types has not always matched theory. Therefore, a scientifically valid process is needed to assess launch vehicle yields considering both propellant characteristics and vehicle characteristics.

1.5. The Importance of Managing Risk. Host sites, supporting a vastly increased launch cadence, are considering risk-based approaches that couple knowledge about potential launch mishap severity with understanding of the possible impact to the public, the environment, and other proximal assets. This process endeavors to build that knowledge for new propellant combinations. As a result of this evaluation, risk acceptors and launch providers can better evaluate the risk to the public, personnel, and surrounding facilities and hardware.

1.6. Relationship to Other Regulations and Directives. U.S.-based commercial companies must comply with all government federal range requirements and the FAA licensing process. The explosive yield methodologies in this manual standardize the testing and analytical framework used to evaluate risks associated with new propellant combinations; however, these methodologies exist in parallel with, and do not address, other applicable regulatory requirements. Final hazard risk acceptance and enforcement are executed through established safety authorities and the FAA licensing process, as applicable.

Chapter 2

KEY CONCEPTS AND CONSIDERATIONS

2.1. Defining Explosive Yield. One of the reasons that predicting equivalent explosive yield in the event of launch vehicle failures is so complex is the unique propellant mixing, ignition, and reaction characteristics. Additionally, there is a significant difference between the explosive results from liquid combinations when compared to standard explosives which have well defined parameters. Another is that, while experience has demonstrated that launch vehicle design and configuration play a large part in the actual yield results, often the launch vehicle variability between launch providers is too great to accurately and reliably predict outcomes.

2.2. Traditional Propellants vs. Novel Propellants .

2.2.1. The results from earlier testing reported in 1968 generated a large body of knowledge about two commonly used rocket propellants: Liquid Oxygen and Rocket Propellant-1 (LOX/RP-1, or LO2/RP-1) and Liquid Oxygen /Liquid Hydrogen (LOX/ LH2, or LO2/LH2).

2.2.2. Real-life incidents and two NASA test campaigns conducted in the 1990s used to simulate launch vehicle explosions called HOVI (Hydrogen/Oxygen Vertical Impact test) and LSHOE (Large Scale Hydrogen/Oxygen Explosion) further refined the empirical knowledge.

2.2.3. If the propellant chemistry that drives mixing and combustion differs from those already characterized, each must undergo some degree of testing to understand the influence of these variables that could lead to unique explosive behavior.

2.2.3.1. For example, the explosive yield determination process is currently being applied, verified and validated to characterize the explosive parameters and ignition phenomena of liquid oxygen (LOX) and liquid methane (CH4 and its derivatives).

2.2.3.2. This propellent combination is unique from LOX/RP-1 and LOX/LH2 in that LOX/LCH4 is highly miscible, meaning both components can exist in the liquid state at similar temperatures and have the chemical potential to create a homogenous liquid mix. However, explosive yield is driven not solely by this chemical miscibility, but also by the physical scale and rate of mixing achieved during a dynamic failure scenario. Any lessons learned from the application of this process will be applied to future updates to this handbook.

2.3. Factors Influencing Explosive Yield, Bounding Cases and Key Design Attributes. The process used to assess launch vehicle yields considers both an in-depth look at propellant characteristics, while also taking a broader view of vehicle characteristics and failure effects. The design process offers opportunities to evaluate bounding cases of failure, with key design features being assessed for mitigating or enhancing explosive yield.

2.4. Main Areas of Study. There are three main areas of study: explosive yield, thermal hazard, and fragmentation risk. Key experimental data include pressure and impulse data to determine the potential explosive yield and enable analysts to produce predictive models. Thermal and visual data anchor models to predict deflagration (fireball) heat rate or temperature, size, and formation. Fragment information, including impact location, velocity, dimensions, weight, and shape classification, informs debris impact risk.

2.5. The Role of Testing and Modeling. Both empirical test data and analysis from engineering models are necessary – they complement each other and help account for the relatively small set of existing test data for other propellant types. Modeling alone cannot provide a sole replacement for testing. The tests described in this manual are largely analogous to legacy tests used to assess scientifically valid yield determinations, however the community now augments the information gleaned from the tests with high fidelity modeling techniques and new methods of recording and analyzing data.

2.6. Historical Context: Project PYRO . The groundbreaking work for improving the definition of the potential hazards of liquid rocket propellant explosions was an extensive experimental Liquid Propellant Explosive Hazards Study in 1963 dubbed Project PYRO. The effort endeavored to find reliable methods to determine the dependence of the blast or shock wave hazards on various test conditions. It also evaluated the heat transfer or thermal hazard of the test explosions.

2.7. Blast Damage Assessment. The explosive yield determinations derived from the process detailed in this handbook serve as a primary input for all blast damage and hazard assessments, both for pre-launch explosives siting and flight safety analysis purposes. These assessments will be conducted using approved Range Safety processes, tools, and models for calculating quantity distance, maximum credible event, risk-based clear zones, expectation of casualty, debris fields, and blast effects (e.g., referencing existing DAF manuals or authorized software tools).

2.8. Other Key Considerations. This process can have many impediments to successfully gaining data to meet desired risk acceptability criteria. The investigation of a novel propellant must weigh the desire to move quickly with the need to understand and manage the risks posed. This process mitigates the necessity of a full complement of explosive tests, which can decrease cost and time commitment, however, investment in tests and modeling that provide meaningful results can still be expensive and time consuming. The evaluation team must have the expertise to uniquely tailor the process to obtain input on the most significant variables affecting the physical phenomena involved. The results must balance theory, models, analysis, and test data for validation and feedback. A comprehensive assessment needs appropriate sponsorship and commitment to devote the resources to arrive at reasonable and reliable conclusions, ensuring appropriate test rigor and availability of subject matter expertise.

Chapter 3

INITIATING A PROPELLANT ASSESSMENT

3.1. Initiation of Propellant Assessment. Review of explosives characteristics of legacy liquid propellants has shown that extrapolation to novel propellants may not apply. Therefore, it is expected that each new or novel propellant assessment must complete this process prior to test or launch on government facilities or use current published government guidance for the specific propellant combination. Any proposed assessment request from a commercial or federal entity should be accompanied with a technical data package that contains a characterization of the propellant combination's readiness or technical maturity, and as much information as is currently known about the propellant and the propulsion concept. Additional information about a proposed vehicle such as the configuration, size, mass ratio, intended mission parameters, failure modes, etc. will inform the timeline and the criticality of developing a robust approach to accommodate the intended usage frequency.

3.2. Assessment Activities. Once initiated, the test team will develop a charter for the effort, draft a program plan, establish success criteria, and provide a cost estimate for the assessment. If the full amount cannot be obtained, the team will advise on programmatic risk mitigation strategies, with appropriate risk evaluations to the launch providers and launch programs. Cooperative development and data sharing with interested industry partners, using cooperative research and development agreements (CRADAs) or other applicable methods such as Space Act Agreements (SAAs) will be utilized to facilitate resource and data sharing to the utmost extent practicable.

Chapter 4

THE THREE-STEP PROCESS: A DETAILED GUIDE

4.1. Process. This chapter details the comprehensive, scientifically valid methodology required to characterize the explosive yield of space launch vehicle propellants.

4.1.1. This chapter should be used as a sequential but tailorable framework. It is not intended to be a rigid checklist; rather, it provides a structured approach that must be customized based on the novelty of the propellant, existing historical data, and the specific launch vehicle configuration, as applicable. Users should evaluate the exit criteria at each step to determine if subsequent steps are necessary or if sufficient data has already been gathered to safely proceed.

4.1.2. This chapter outlines the lifecycle of an explosive yield determination campaign, from initial theoretical research and computational modeling, through the execution of physical explosive testing (at lab, small, and large scales), to the final refinement of predictive hazard models used for operational safety approvals. The three steps function as an iterative, progressive, and complementary system.

4.1.2.1. Step 1 (Theoretical-Analytic Review): Establishes the baseline understanding of the propellant's chemical and thermodynamic properties, identifying bounding failure scenarios and worst-case predictions to inform test planning.

4.1.2.2. Step 2 (Characterization and Bounding Tests): Executes physical testing to empirically challenge and validate the theories developed in Step 1. This step anchors computational models with real-world pressure, thermal, and fragmentation data.

4.1.2.3. Step 3 (Refinement Tests): Acts as the final tuning phase. It specifically targets unresolved anomalies, data gaps, or highly specific use-cases identified during Step 2, ensuring the final predictive models are robust and ready for operational use.

4.2. Step 1: Theoretical-Analytic Review and Analysis.

4.2.1. Purpose. The first step of the process, theoretical-analytic review, entails a review of the information known of the chemistry, mixing, and explosion characteristics of the propellants of interest.

4.2.2. Activities.

4.2.2.1. Conduct a literature search. Research the history of the individual propellants and propellant combination, any documented applicable accidental or test explosion data that might be used as a starting point, and any applicable theoretical, numerical, or empirical models.

4.2.2.2. Characterize the propellant.

4.2.2.2.1. Basic properties of the components (molecular formula, density, melting and boiling points, heat of formation, latent heat of vaporization, viscosity, specific heat, ratio of specific heats, and thermal conductivity).

4.2.2.2.2. Physical properties of the propellant combination (density, miscibility, homogeneity, likelihood of cavitation, melting and boiling points, heat of formation,

latent heat of vaporization, viscosity, specific heat, ratio of specific heats, and thermal conductivity, etc.).

4.2.2.2.3. Ignition phenomena (hypergolic or non-hypergolic, ignition mechanisms and thresholds).

4.2.2.2.4. Thermochemical (adiabatic flame temperature, specific heat, combustion products, etc.).

4.2.2.2.5. Other pertinent information: can each component and the resulting mixture be assumed to behave as an ideal gas, are the vapors from unreacted materials or materials of combustion toxic, is the combination sensitive to external factors such as weather, are there any instabilities noted, is the obtainable purity sufficient, does the propellant decompose or coke, is it compatible with surrounding materials, etc.

4.2.2.3. Characterize mixing and explosive properties. The explosive characterization of the propellants will inform the areas of greatest risk when evaluating failure modes associated with space LVs: the basic premise is that only the fuel and oxidizer that mix with each other can be a part of the explosion. Studying the kinetics of the possible or proposed configurations, knowing their thermophysical properties and how they might mix together, change phase, or ignite is crucial to assessing the mixing and explosives properties of the propellants of interest. In general, the closer the mixture is to the stoichiometric composition, the stronger a detonation will be. Any failure that results in a well-mixed system that could be detonated by a strong shock or self-ignition will often be the "worst case" scenario, or strongest explosion.

4.2.2.4. Predict the "worst case". The intent of this action is to obtain knowledge of the interaction between the chemical components based on the miscibility, scalability, ignition, vaporization, dispersion, autoignition, post-detonation reactions, and other pertinent energetic behaviors for the propellant combination as well as those of the products of their combustion. Depending on the energetic materials involved, the credibility of this prediction could be influenced by a given system's operating and design parameters.

4.2.2.5. Review applicable analogous testing. In some cases, enough information may be known about the liquid rocket propellant combinations to enable helpful thermochemical simulation through means like computational fluid dynamics (CFD) modeling or finite element analysis (FEA). In other cases, little may be known about the interaction and substitutes with similar properties may have to provide the starting point for expert judgements about the driving variables. Testing with substitutes can help identify modeling parameters to characterize the explosive potential of a liquid propellant combination.

4.2.3. Expected Outcomes. The outcome is an informed and tailored test matrix for characterization discovery and bounding tests. If this theoretical review and analysis activity demonstrates that the full scope of testing is not necessary, it may be appropriate to skip the next major step in the explosive yield characterization process, establishing a test program from scratch. Instead, proceed to Step 3, Refinement Tests, to explore a particular regime, use case, unknown phenomena, or to reduce conservatism of the information already known.

4.2.4. Exit Criteria. The conclusion of the theoretical-analytic process will provide a better understanding among subject matter experts on bounding tests deemed necessary and acceptable to safety authorities. This provides a reasonably conservative estimate of the risk,

using a probabilistic approach that can be translated into explosive equivalence or other models to inform safety standards. The subject matters experts will then generate a recommended bounding test matrix and work with supporting and test team personnel to create a cost and time estimate to complete, as well as any recommended refinement testing that might be considered as follow-on.

4.3. Step 2: Characterization and Bounding Tests.

4.3.1. Purpose. The objective of characterization and bounding tests is to identify the scenarios that produce the most reasonably foreseeable harmful conditions to people and critical assets surrounding the launching system, thereby anchoring theoretical models and validating expected explosion characteristics to inform risk assessments. From step 1 we have the worst-case prediction, but bounding tests should be established to smartly design tests to what is credible and achievable. It is important to note, although not addressed here, that hazards could result from more than the blast and shock wave of an explosion and fragmentation, such as chemical toxicity and radiation. The goals of these tests are to primarily:

4.3.1.1. Produce and collect pressure and impulse data necessary to determine the potential explosive yield or to produce an alternative predictive model. **Note**: Additional factors of consideration that can affect pressure and impulse data include but are not limited to: height-of-burst, geometric shape variation, and surface burst strengthened due to ground reflection.

4.3.1.2. Collect data, including thermal or visual, to establish fireball size and formation.

4.3.1.3. Collect any data that could be relevant to far field blast overpressure modeling, such as weather profiles and very far field pressure and impulse measurements to establish attenuation trends.

4.3.1.4. Collect and document fragment information, including impact location, dimensions, weight, and shape classification.

4.3.1.5. Test as closely to expected flight-like conditions as reasonably possible, including dimensions, scale, mixture ratio, geometries, pressure, pressure differentials, and temperatures.

4.3.1.6. Produce statistically significant results for small-scale tests.

4.3.1.7. Perform multiple large-scale tests to establish confidence in trends and scalability factors.

4.3.1.8. Ensure that the credible explosive yield potential is enveloped under both static (e.g., during propellant fill and launch delays) and dynamic conditions (e.g., high-velocity impacts).

4.3.2. Types of Tests. Laboratory-scale tests, small-scale characterization tests, and large-scale testing.

4.3.2.1. The laboratory and small-scale tests focus on understanding the mixing and ignition phenomenon to inform theoretical models. They typically vary ignition source, ignition delay, mixture ratio, and fuel composition to examine mixing behaviors and explosive yield. This may be crucial for propellants in cryogenic/liquid and/or gaseous phases.

4.3.2.2. The smaller scale propellant reaction characterization tests attempt to establish bounding or driving factors related to explosive yield, thermal hazard, and fragmentation risk before planning larger, more realistic (e.g. flight-like) and expensive tests. The resultant data inform physics-based and engineering models and produce a tailored test matrix to focus larger test efforts on the most significant attributes requiring evaluation at that scale. Key experimental data gathered for the larger scale tests include pressure and impulse data to determine the potential explosive yield and update predictive models. Experience shows that data relevant to far field blast overpressure modeling, such as weather profiles and very far field pressure and impulse measurements, can establish attenuation trends. Fragment information, including impact location, velocity, dimensions, weight, and shape classification, informs debris impact risk.

4.3.2.3. Historically, scalability has only been determined empirically; therefore, it is desirable to perform some large-scale tests as close to expected flight-like conditions as reasonably possible. If not possible or practical, conservatism is employed. This could be done by adding a safety factor or not extrapolating a decaying yield curve/analytical model to vehicle masses past those with actual data.

4.3.3. One challenge is that there is no one “yield” for the entire set of conditions experienced during a launch lifecycle. Pre-flight conditions during fueling can vary significantly over time. Conditions during flight could be entirely different than those prior to lift-off. Assumptions may not be valid throughout propellant fill, tank pressurization, ignition, burning, shock propagation, etc. Combustion may have varying effects and the methods of characterizing these effects may differ. Failure mode can have a significant impact, so it is difficult to establish a truly generic model for many propellant combinations. Therefore, specific testing should be tailored to the propellants and type of reaction, the general aspects of the systems under consideration that might affect the ultimate yield, the damage mechanism and other impact factors like speed, surface, and orientation of impact. Many of the variables to consider are listed below:

4.3.3.1. Propellant amount (fuel to oxidizer ratio).

4.3.3.2. Tank pressure.

4.3.3.3. Pressure differential between the tanks.

4.3.3.4. Tank length to diameter ratio (L/D).

4.3.3.5. Contact area.

4.3.3.6. Propellant temperature.

4.3.3.7. Tank configuration.

4.3.3.8. Feed tube configuration.

4.3.3.9. Tank orientation.

4.3.3.10. Tank ullage.

4.3.3.11. Ignition source.

4.3.3.12. Ignition time.

4.3.3.13. Mixing time.

4.3.3.14. Other factors:

4.3.3.14.1. Tank composition.

4.3.3.14.2. Sub-cooling effects.

4.3.3.14.3. Propellant composition.

4.3.3.14.4. The potential for sympathetic detonations.

4.3.3.14.5. Effect of impact surface variability.

4.3.3.14.6. Contributory explosive effects caused by initiation of installed ordnance components.

4.3.4. Lab Scale Testing.

4.3.4.1. The first approach for any new combination of propellants or novel vehicle design will be to perform small scale or lab scale testing, due to the relative ease of conducting multiple tests to challenge and validate assumptions at this scale. Several assumptions and decisions must be made at this step to attempt to ensure continuity to larger scale testing.

4.3.4.1.1. The influence of the composition of the tested propellants needs to be evaluated to ensure purity relevant to that obtainable in larger quantities.

4.3.4.1.2. The measurement methodology needs to be evaluated such that it is translatable and informative to models and to larger scale testing.

4.3.4.1.3. Conditions at lab scale are presumably more rigidly controlled than at large scale, so the influence of weather and other external effects should be investigated, if possible.

4.3.4.1.4. Anything that might not be easily testable at larger scales should be investigated at lab scale.

4.3.4.1.5. Testing should be done to isolate variables to determine if the difference is significant or negligible with the propellants involved.

4.3.4.2. Lab scale testing should include, at a minimum:

4.3.4.2.1. Characterization of the propellant reaction, the products of combustion and identify conditions conducive to a detonation scenario.

4.3.4.2.2. Investigation of mixing characteristics at the worst-case propellant state (temperatures and pressures) or the foreseeable range of propellant states if the worst case is uncertain.

4.3.4.2.3. Testing of mixture ratio effects on yield.

4.3.4.2.4. Investigation of ignition source and effect upon yield.

4.3.4.2.5. Tests that attempt to ascertain the optimal test matrices for larger scale and impact tests.

4.3.4.2.6. Any other test that might isolate variables or further refine knowledge of scalability, hazard dependencies, mixing turbulence, secondary detonation, impact reactions, etc.

4.3.5. Small-scale Characterization Testing.

4.3.5.1. Small-scale characterization testing will acquire data on the explosive characteristics of rapidly mixed (i.e., large surface contact area) commodities under controlled propellant conditions, amounts, and configurations, that cannot be performed at lab-scales, to:

4.3.5.1.1. Measure the near and mid-range blast overpressure and impulse.

4.3.5.1.2. Measure the thermal characteristics of the fireball.

4.3.5.1.3. Measure the time to spontaneous ignition (or control the time to forced ignition).

4.3.5.1.4. Acquire high-speed video data on the characteristic size, shape, and growth rate of the fireball.

4.3.5.1.5. Test the theoretical energetic yield limit. This is helpful data to evaluate near-field risk assessments for crew, payload, and nuclear safety purposes.

4.3.5.2. One known test method for propellant mixing characterization is the Propellant Reactivity Characterization (PRC) Test (also known as a Distributed Mixing Test) – multiple glass dewars (insulated containers) are filled with one propellant and placed in a pan filled with the other propellant. A detonation under the pan floor shocks the dewars and shatters the glass, creating a mixing interface area. The propellant mixture is then ignited, providing information about the impact of ignition delay, mixture ratio, and fuel composition.

4.3.6. Large Scale Testing.

4.3.6.1. Large scale testing must be done to allow for insight into both structural and fluid dynamics to actual launch vehicles. The test team must meticulously plan these tests to ensure maximum information can be gleaned from what will likely be a limited set of tests to explore the considerable variability that may be found, particularly if the blast phenomena are difficult to reproduce exactly.

4.3.6.2. If tests cannot be conducted at actual launch vehicle sizes, test planners should develop a defensible alternative scaling strategy. Test organizations must formally document their scaling methodology, clearly identifying the limitations of the sub-scale tests, the resulting data gaps, and how those gaps will be mitigated (e.g., via conservative modeling or safety factors). This strategy, which might involve a mid-scale test regime, analogous tests, or utilizing an appropriately instrumented first flight to capture data in the event of a mishap, must be reviewed for sufficiency by subject matter experts and approved through normal risk acceptance channels.

4.3.6.3. Impact tests. These are tests that imitate a failure mechanism such as a fallback from a high altitude or powered flight impact. Typically, the propellant tank(s) is dropped from a vertical height and allowed to impact a target. Propelling a test article along a horizontal track to simulate impact is technically feasible but is considered a non-preferred methodology, as the horizontal acceleration can introduce non-flight-like fluid sloshing and structural pre-loading that complicates yield determination. Historically used versions of these tests, intended to test different failure modes are:

4.3.6.3.1. Dynamic Confined by Ground Surface (CBGS). Both oxidizer and fuel are released at nearly the same time and velocity by dropping a test article filled with each separated by a frangible common bulkhead. The drop height can be varied or an acceleration device added to obtain different velocities. Tanks are dropped onto a cutter on the ground surface, the bulkhead and downcomer shatter and the tank walls rupture. Propellants then mix, with autoignition likely. In these cases, tests provide valuable information on the function of impact velocity on autoignition, mixing times, and overall yield.

4.3.6.3.2. Static CBGS. Oxidizer and fuel are released at different times and velocities, spilling on the ground, and then ignited. The spilled propellants can be confined within a shallow basin or within a silo. The impact velocity of the fluids is generally low.

4.3.6.4. Confinement by the Missile (CBM). Tests that imitate an internal failure where the propellants remain confined by the walls of the missile. In these tests a diaphragm between fuel and oxidizer tanks is ruptured and all mixing takes place within the propellant tank. This provides information regarding ignition delay, propellant mass effects, the influence of tank orientation, contact area, tank pressure, etc.

4.3.6.5. Sympathetic Detonation. A sympathetic detonation test may be conducted using a solid motor stage or segment in near intimate contact with the liquid propellant test article. The liquid article used will be chosen based on the CBM/CBGS tests and the yield from that test will be used as the baseline to determine the solid motor contribution to the composite yield. The test plan will need to specify details of the test configuration, stage/motor orientation, apparatus design, instrumentation, test operations, and data analysis.

4.3.7. Instrumentation. It is essential that instrumentation is available and calibrated to collect optimal data on the behavior of the blast wave, overpressure, asymmetry, thermal and fragmentation effects, etc.

4.3.7.1. Instrumentation will notionally consist of blast pressure gauges, thermocouples, radiometers, film and video cameras (low and high speed), and ambient weather instrumentation. Ensure all instrumentation have a current calibration certification and appropriate thermal protection.

4.3.7.2. Thermal instrumentation may include radiometers, thermocouple probes, and reference material slabs to measure heat flux both inside and outside the fireball. Thermal data may also be collected via thermal cameras or other measurement analysis techniques.

4.3.7.3. The engineering organization must ensure all measurements are traceable to the test agency’s calibration standards and report this uncertainty as an integral part of the final data package.

4.3.7.4. For most large-scale test configurations, the external sensor and gauge location distances from ground zero will increase with increasing propellant weight.

4.3.7.5. The testing organization should ensure the instrumentation does not directly affect the test results. For example, instrumentation should not cause premature explosions, limit propellant reactions, influence mixing, etc.

4.3.7.6. Test organizations are highly encouraged to precede involved, large-scale tests with a baseline calibration shot using a known high-explosive standard, e.g., trinitrotoluene (TNT) or Composition C-4. This known energy source serves as a reliable reference point to ensure the instrumentation suite is properly calibrated and capturing data as expected. This practice allows testers to troubleshoot sensor arrays, mitigate data-loss risks, and gain preliminary environmental insights before committing resources to the full-scale propellant tests.

4.3.8. Expected Outcomes. The expense of each test will likely limit the test team's ability to conduct repeat tests. Thus, models and yield curves may be expected to include a degree of conservatism reflective of that difficulty. Test planners and management should assign and accept risks to this concern, addressing consequences and mitigations, especially in the event that test objectives may not always be initially met and must be addressed later during refinement testing.

4.3.9. Exit Criteria. The characterization and bounding test phase is considered complete when empirical testing has sufficiently validated the theoretical hazard models and established definitive bounds for subsequent risk assessments. Specifically, the phase concludes when the following criteria are satisfied as determined collaboratively by the subject matter experts, testers, and safety authorities:

4.3.9.1. Data Sufficiency. Required pressure, impulse, thermal, and fragmentation data from lab, small-scale, and large-scale tests have been successfully collected, calibrated, and documented without critical loss of fidelity.

4.3.9.2. Model Validation. Predictive models for explosive yield, fireball formation, and blast overpressure have been updated and anchored against the empirical test data.

4.3.9.3. Scenario Quantification. The most reasonably foreseeable failure scenarios—across both static (e.g., fueling) and dynamic (e.g., flight impact) conditions—are mathematically bounded to accurately inform personnel and critical asset risk assessments.

4.3.9.4. Uncertainty Mitigation. Any remaining test variables or gaps in scalability data (due to the limitations of reproducing full-scale flight vehicles) have been formally accounted for through documented safety margins or conservative analytical assumptions.

4.4. Step 3: Refinement Tests.

4.4.1. Purpose. The third step of the process, refinement testing, explores any unique considerations, areas of higher uncertainty in the results, or gaps or risks within the tests already conducted. These refinement tests can consist of lab scale testing, larger scale testing, both, or specialized testing intended to focus on a particular failure mechanism or phenomenon.

4.4.2. Although generally performed after the other two process steps, it is important to note that refinement testing may occur in conjunction with the second step or even performed years after initial testing of the propellant. It is separated out in the interest of avoiding the need to create an "all-encompassing" test series for the second step and allow planning for examination of unexplained phenomena that may arise or testing at higher fidelity without overcomplicating an ongoing test campaign.

4.4.3. Incorporation of Industry/User Results. Incorporation of industry or user test results, analysis, or support may be vital to address a unique failure mode or phenomenon of concern.

4.4.4. Exit Criteria. The refinement testing phase is considered complete when targeted investigations have sufficiently resolved identified data gaps, and the resulting hazard models are formally approved for operational application. Specifically, the phase concludes when the following criteria are satisfied as determined collaboratively by the subject matter experts, testers, and safety authorities:

4.4.4.1. Anomaly and Gap Resolution. Specific data gaps, unacceptable levels of uncertainty, or unexplained phenomena identified during initial characterization testing (Step 2) have been successfully investigated and resolved through targeted testing or analysis.

4.4.4.2. Variable Integration. The refined explosive yield models accurately and dynamically account for critical mission variables, including propellant quantities, specific failure modes, impact velocities, and other defined characterization metrics.

4.4.4.3. External Data Synthesis. Where applicable, relevant industry or user test results, analyses, and support data have been formally evaluated, validated, and incorporated to address unique failure modes.

4.4.4.4. Operational Application Readiness. The finalized models are explicitly formatted and ready to directly inform concrete operational decisions, including explosive siting, range safety protocols, mission-specific risk assessments, and hazard mitigation strategies.

Chapter 5

CONCLUSION

5.1. Key Takeaways. This handbook provides the Blast Damage Assessment Guide and a jointly established and scientifically valid process to produce information required to assess explosive safety risks for space launch vehicles. It was produced in collaboration with NASA and the FAA Office of Commercial Space. It documents a response to the FY26 NDAA, Section 1610, and FY24 Section 1601.

5.2. Process Summary. The process describes a three-step framework to conduct exploration into the explosive properties of a propellant combination under study to observe and model phenomena pertinent to its explosive behavior under foreseeable vehicle configurations and failure modes. The process will also result in an understanding of the explosive yield, thermal hazard, and fragmentation hazards of a space launch vehicle. The derived information will inform risk analysis for multiple applications, including flight safety analysis, space launch related explosives siting, launch of radioactive materials, and human spaceflight.

Chapter 6

ADDITIONAL TECHNICAL RESOURCES

6.1. Recommendations for Further Reading. Users of this handbook may benefit from additional research and publications on the topic of explosive yield characterization.

6.1.1. Sutton et al, Synopsis of LOX/Liquid Methane & Liquid Natural Gas Rocket Propellant, AIAA 2019-4283, Published Online on 16 Aug 2019 **https://doi.org/10.2514/6.2019-4283**

6.1.2. Berkowitz and Titulaer, Development of a Lox/Methane Intact Impact Yield Curve for a New Launch Vehicle, Journal of Space Safety Engineering, Volume 11, Issue 4, Dec 2024, pp. 590-604. Available at **https://www.sciencedirect.com/science/article/pii/S2468896724001770**

6.2. Disclaimer. The intent with this list is to share previously published material on the subject and acknowledge contributions made by experts in the field but does not constitute endorsement of the specific processes and methods contained within.

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

DAFI 91-202, *The Department of the Air Force (DAF) Mishap Prevention Program*, 20 March 2020

PL 118-31, *National Defense Authorization Act for Fiscal Year 2024, Section 1601, Delegation of Certain Authority of Explosive Safety Board*, 22 December 2023

PL 119-60, *National Defense Authorization Act for Fiscal Year 2026, Section 1610, Blast Damage Assessment Guide for Space Vehicles at Air Force Launch Complexes*, 18 December 2025

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Product*

Abbreviations and Acronyms

CBGS—Confined by Ground Surface

CBM—Confinement by the Missile

CFD—Computational Fluid Dynamics

CH4—Methane

CRADA—Cooperative Research and Development Agreement

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DOT—Department of Transportation

FEA—Finite Element Analysis

FAA—Federal Aviation Administration

FY—Fiscal Year

HOVI—Hydrogen/Oxygen Vertical Impact test

L/D—Length to Diameter Ratio

LH2—Liquid Hydrogen

LCH4—Liquid Methane

LO2—Liquid Oxygen

LOX—Liquid Oxygen

LSHOE—Large Scale Hydrogen / Oxygen Explosion

LV—Launch Vehicle

NASA—National Aeronautics and Space Administration

NDAA—National Defense Authorization Act

OPR—Office of Primary Responsibility

PRC—Propellant Reactivity Characterization

RP-1—Rocket Propellant-1

SAA—Space Act Agreement

SecAF—Secretary of the Air Force

SME—Subject Matter Expert

SSC—Space Systems Command

TNT—Trinitrotoluene

USAF—United States Air Force

Office Symbols

SSC/SE—Space Systems Command, Directorate of Safety

SSC/SEK—Space Systems Command, Space Safety Division

Terms

Deflagration—A subsonic exothermic reaction characterized by a flame front propagating through a material.

Detonation Reaction—A supersonic decomposition reaction propagates through the energetic materials and produces an intense shock in the surrounding medium and very rapid plastic deformation of metallic cases, followed by extensive fragmentation. All energetic materials will be consumed. Effects will include large ground craters for items on or close to the ground; holing, plastic flow damage, and fragmentation of adjacent metal structures; and blast overpressure damage to nearby structures.

Explosive Yield—A measure of the energy released in an explosion, measured in terms of energy released per mole or per mass. It is common to express explosive yield in terms of the equivalent amount of the chemical explosive, trinitrotoluene (TNT). However, liquid propellant explosions are generally not well-characterized in terms of TNT equivalence, at least in the near-field, where a different amount of TNT would provide an equivalent overpressure than would provide an equivalent impulse.

Hypergolic—A property of various combinations of chemicals to self-ignite upon contact between the fuel and oxidizer, without a spark or other external initiation source.

Miscibility—The ability of two or more substances to completely dissolve and mix in all proportions, creating a homogeneous mixture.

Non-Hypergolic—Non-hypergolic propellants require an ignition source (e.g., spark, flame) to initiate combustion.

Overpressure—The pressure exceeding normal atmospheric pressure caused by a shock wave from an explosion.

Propellant—A chemical substance or mixture of substances that is burned to produce thrust in a rocket or other projectile.

Stoichiometric Composition—The ratio of fuel and oxidizer that results in complete combustion, with neither reactant in excess.

Space Flight Participant—An individual, who is not crew or a government astronaut, carried on board a launch vehicle or reentry vehicle. (see 14 CFR 401.7)

Thermodynamically Similar—Having similar thermodynamic properties, such as heat capacity, heat of combustion, and heat of vaporization, which influence their explosive behavior.

Ullage—The unfilled space in a propellant tank, which can affect mixing and other factors.